Impact of Sea Ice Melt on Chlorophyll Concentrations in the Labrador Sea

In recent years, rising temperatures and climate change have led to an accelerated melting of sea ice, raising concerns about its effects on marine life. One of the key impacts is on the ocean's primary productivity, as the melting of sea ice may influence the distribution of chlorophyll, a vital indicator of phytoplankton growth.

Multiple complex physical, chemical, and biological variables influence phytoplankton concentrations across the Arctic.

This presentation will explore the relationship between sea ice and the phytoplankton levels over the Labrador sea using remotely-sensed chlorophyll data.

Goli Prajwalitha (University of Bremen)

Motivation & Hypothesis

Why the Labrador Sea?

1. Covered by first-year ice (confirmed via OSI SAF Quicklook).
2. Key area for ocean-atmosphere interactions and biological productivity.

Hypothesis:

As sea ice melts, chlorophyll concentrations will increase due to enhanced light availability and nutrient mixing.

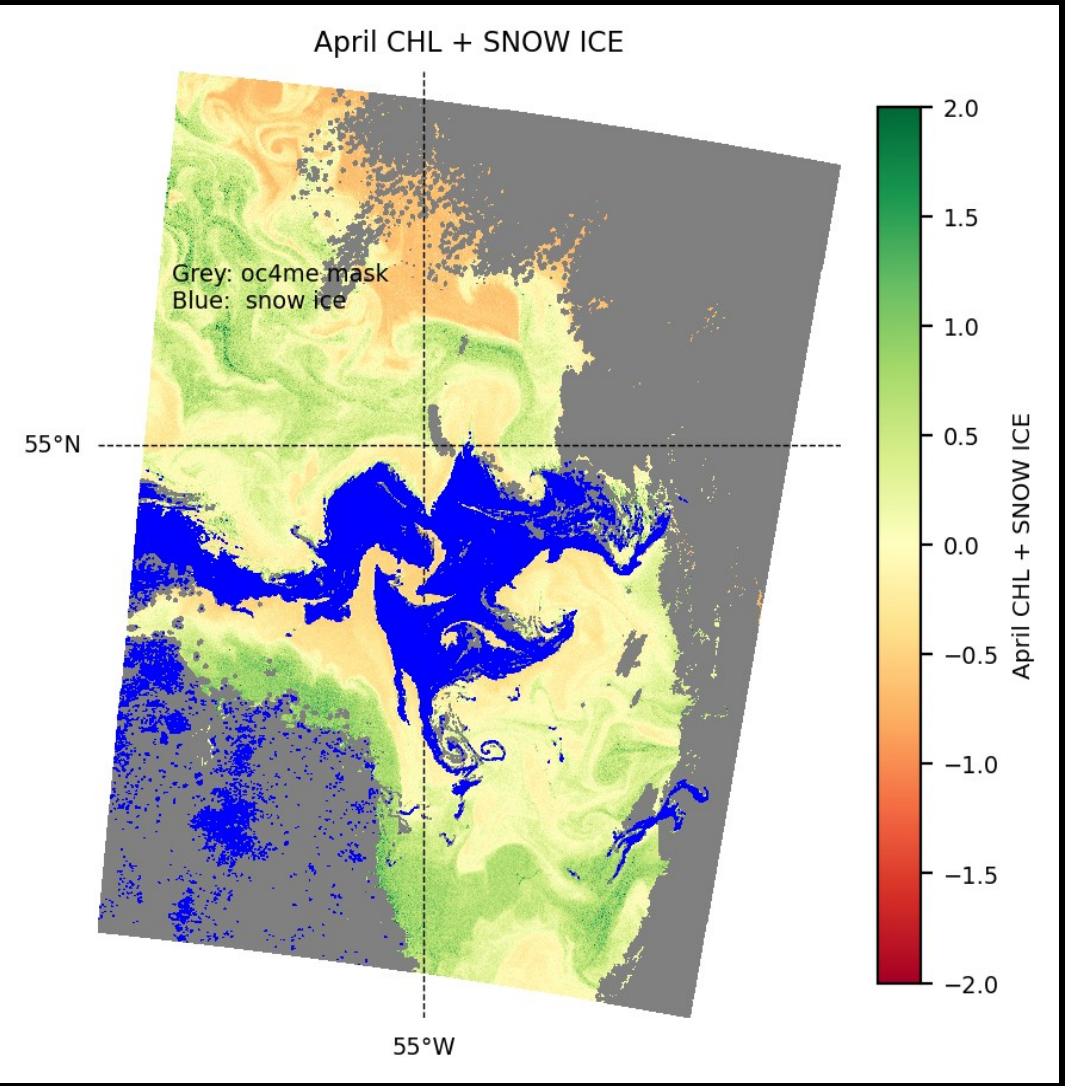
Data and Methodology

Data downloaded from WEkEO

1. **Dataset ID:** EO:EUM:DAT:SENTINEL-3:OL_2_WFR___.
2. **Chlorophyll-a Data:** Measured in mg/m³ using Sentinel-3 OLCI (OC4ME).
3. **Sea Ice Identification:** Using 'SNOW ICE' flags from Sentinel-3.
4. Visualizing and analyzing the data within a **Jupyter Notebook** environment.

Analysis: Comparison of chlorophyll-a concentrations between April 27th and May 27th.

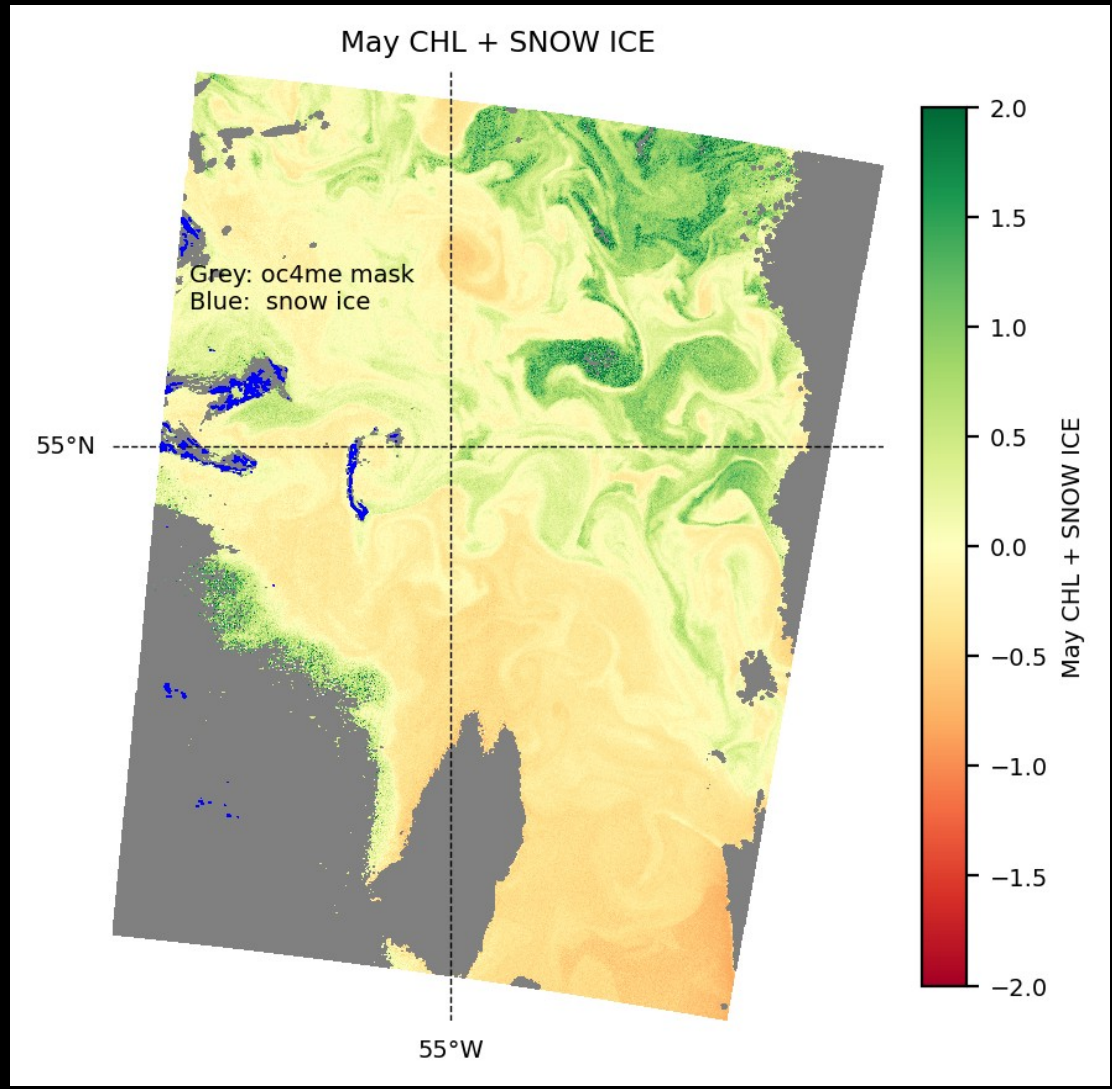
Results



April 27th:
Lower chl-a concentrations

May 27th:
Increased chl-a concentrations

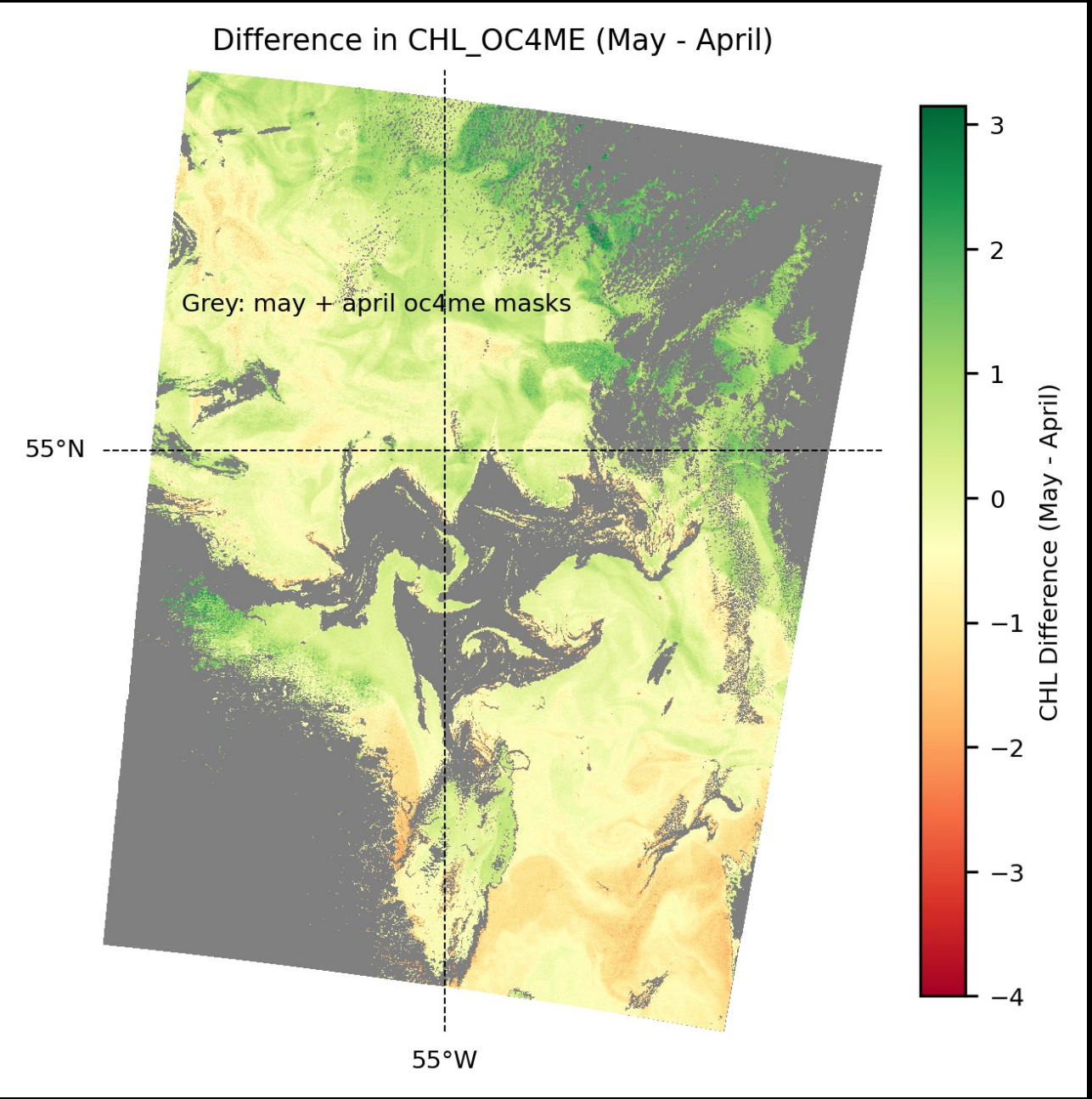
Potential Variations:
Influence of ocean currents, and other environmental factors.



Difference Map

1. The difference map was generated to assess the spatial changes in chlorophyll concentrations over time.

2. The analysis revealed **overall positive values**, indicating a **general increase in chlorophyll concentrations** in areas affected by sea ice melt.



Validation

L4 data from CMEMS

Conclusion & Future Work

Key Takeaway:

Sea ice melt significantly impacts oceanic biological productivity.

Next Steps:

Extend analysis to more time periods for better seasonal trends.

Potential Applications:

Understanding climate change impacts on marine ecosystems.

References

[1] https://user.eumetsat.int/resources/case-studies/sea-ice-swirls-in-the-skagerrak-strait

[2] https://de.m.wikipedia.org/wiki/Datei:Labrador_sea_map.png

[3] https://wekeo.copernicus.eu/

[4] https://osi-saf.eumetsat.int/products/sea-ice-products

[5] https://marine.copernicus.eu/access-data

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